

JETHARAM BHAMBHU

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RESEARCH INTERESTS

Approximation Algorithms, Structural Algorithm Design, Submodular Optimization, Linear Programming and Polyhedra, Limits of LP-based Approximation.

EDUCATION

Indian Institute of Science Education and Research (IISER), Pune
BS–MS Dual Degree in Mathematics

Aug 2021 – Present

RESEARCH EXPERIENCE

Research

Master's Thesis: Submodular Interval Stabbing Problem

May 2025 – Present

Indian Institute of Technology (IIT), Delhi | Supervisors: Prof. Amit Kumar, Prof. Naveen Garg

- Studied approximation algorithms for the Submodular Interval Stabbing Problem (SISP).
- Established APX-hardness for a restricted subclass via polynomial-time reduction.
- Derived an 8-approximation algorithm for the Multilevel Aggregation Problem with Holding and Delay Costs via LP rounding.

Asymmetric Graph Parameters

Jun – Aug 2024

Polymath Jr. REU | Supervisors: Puck Rombach, Daniel P. Johnston

- Investigated the asymmetric coloring number of disconnected graphs.
- Derived and verified this parameter for several graph families, including paths, cycles, and hypercubes.
- Contributed to open graph-theoretic problems in a collaborative research setting.

Reading Projects

Linear Programming and Combinatorial Optimization

IISER Pune

Supervisor: Prof. Prafullkumar Tale

- Studied LP duality, simplex and ellipsoid methods, and applications to approximation algorithms.

Comparability Graphs

IISER Pune

Supervisor: Prof. Soumen Maity

- Studied structural properties of comparability graphs; reconstructed proofs of polynomial-time algorithms for maximum weighted clique.

Isoperimetric Problem

IISER Pune

Supervisor: Prof. Anisa Chorwadwala (Summer Project)

- Reconstructed classical proofs using geometric and variational arguments.

MANUSCRIPTS

The Submodular Interval Stabbing Problem — Master's thesis manuscript.
Approximation Algorithms for Multilevel Aggregation with Delay and Holding Costs.

TALKS

Turing's 1936 Paper: On Computable Numbers

IISER Pune

Pólya Mathematics Journal Club. Expounded Turing's foundational construction of the universal machine and the proof of the undecidability of the Entscheidungsproblem.

Greedy Algorithms and Matroids

IISER Pune

Pólya Mathematics Journal Club. Presented the matroid intersection framework and how the greedy algorithm yields optimal bases.

RELEVANT COURSEWORK

Pure Mathematics: Real Analysis, Functional Analysis, Differential Geometry, Abstract Algebra.
Algorithms & Optimization: Graph Theory, Algorithms, Discrete Mathematics, Linear Programming, Computational Number Theory, Probability.

SKILLS

Programming: Python, Fortran, C++ **Tools:** $\text{\LaTeX}$, Git
Languages: English (TOEFL 108/120), Hindi (Native)

HONORS

KVPY Fellowship 2021 – 2026
awarded to the top 1% of candidates in the national KVPY examination.

JEE Main & Advanced 2021
Among the top 0.3% of candidates nationally in both examinations.

LEADERSHIP

Organizer, Axiomathica Aug 2024 – Present
IISER Pune Math Club. Organized a proof-based mathematics competition; curated problems focused on rigorous reasoning and managed event logistics.

REFERENCES

Professor Amit Kumar
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Professor Naveen Garg
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